Drivers of Forest Loss Across Countries

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Introduction

Research question: Does the impact of industrial activity on ecological damage depend on economic development?

- **Dataset:** Yale Environmental Performance Index (EPI)
  - Combines raw data and synthesized variables in categories of ecosystem vitality, environmental health, and climate change for 220 countries from 1995-2022
- **Our approach:**
  - Identify indicators of industrial activity, economic development, and ecological damage
  - Fit linear regression models using highest quality variables to a training dataset
  - Assess how well the model generalizes on a test dataset

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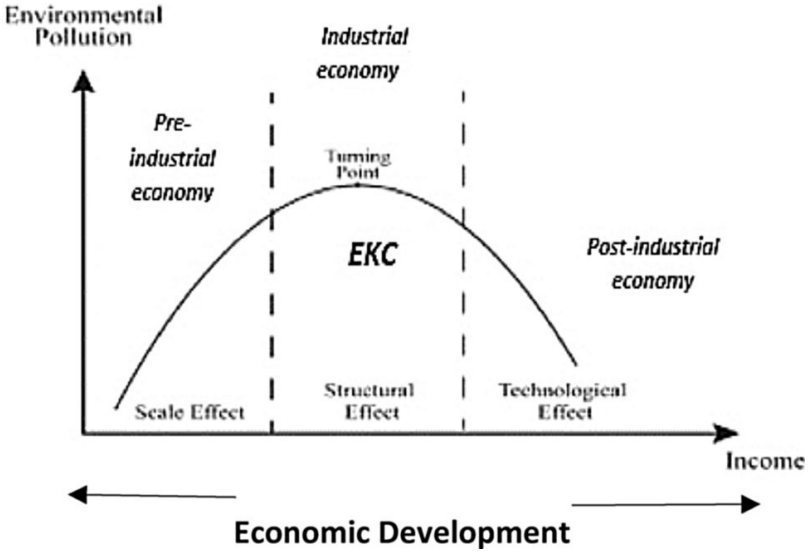
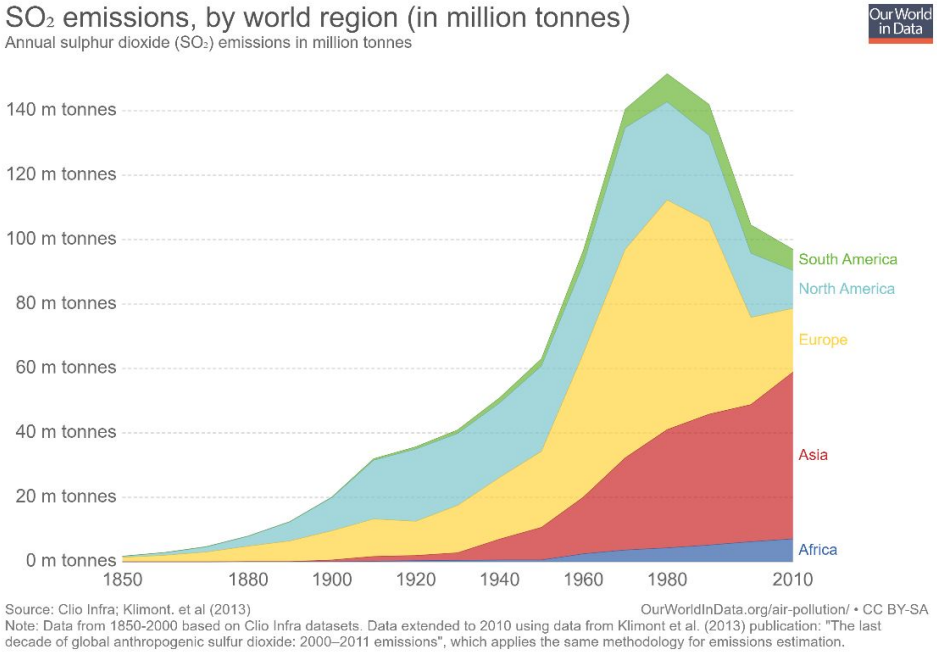
Background research

Research question: Does the impact of industrial activity on ecological damage depend on economic development?

The **Environmental Kuznets Curve (EKC)** is a theorized relationship between these factors





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Relevant variables in the dataset

Task 1: Identify indicators of industrial activity, economic development, and ecological damage from EPI dataset

Ecological Damage (Response Variables)

- **FCL** – Forest cover loss (~ forest loss adjusted for how permanent the deforestation is)
- **PFL** – Primary forest loss (~ loss of forests as a proportion of original forest area)
- **FSS** – Fish stock status (~ weighted percentage of total catch coming from collapsed fish stocks)

Industrial Activity (Predictors)

- **CDO** – CO_2 emissions (raw data)
- **SO2** – Sulfur dioxide emissions (raw data)

Economic Development (Moderators)

- **GDP** – Gross domestic product (raw data)
- **POP** – Population size (raw data)
- **SNM** – Sustainable nitrogen management (efficiency of nitrogen fertilizer usage, potentially indicates maturity of infrastructure)
- **HPE** – Anthropogenic (human) PM2.5 exposure (potentially indicates cleanliness of industry)

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Dataset details

- Dataset contains **220 rows** (countries) and **4966 columns** (variables*years); significant missing data
- Selected years with highest **data completeness** as our training/testing datasets
  - Used full years as training/testing sets due to limited sample size
  - Temporal dependence is a concern with this approach
  - Decided on **2012** and **2021** as train/test years
- Certain variables had high numbers of missing rows (FSS, PFL); did not pursue further
  - FSS (fish stock status) also did not contain any data for inland countries

Category	**Variable**	**Description**	**Missing rows**
Industrial activity	CDO	CO2 emissions	10
Industrial activity	SO2	SO2 emissions	12
Economic development	GDP	Gross domestic product	22
Economic development	HPE	Anthropogenic PM2.5 exposure	22
Economic development	SNM	Sustainable Nitrogen Management Index	26
Other	POP	Population	10
Ecological damage	FCL	Forest cover loss	70
Ecological damage	FSS	Fish stock status	102
Ecological damage	PFL	Primary forest loss	137

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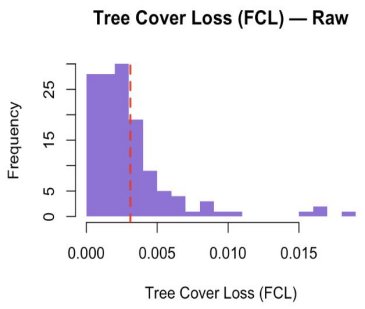
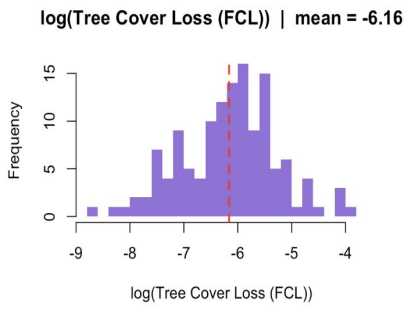
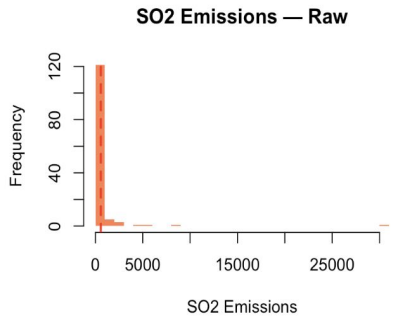
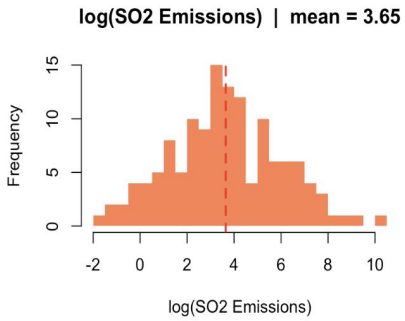
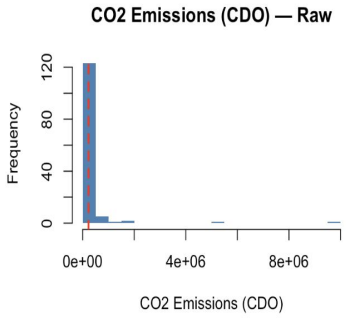
Model Preparation - Choosing raw vs log distributions

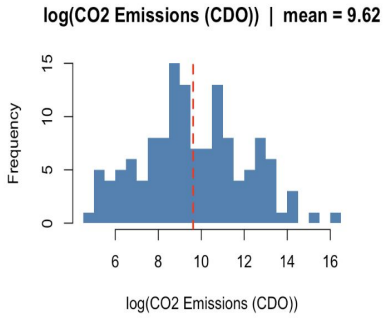


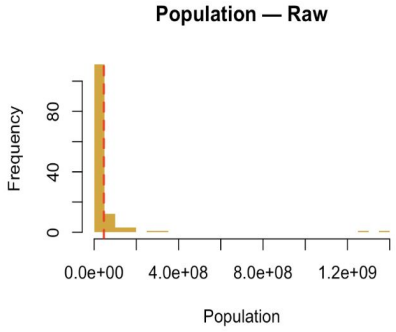


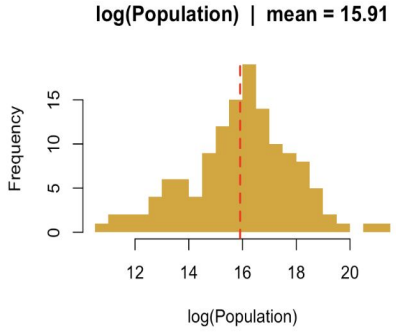












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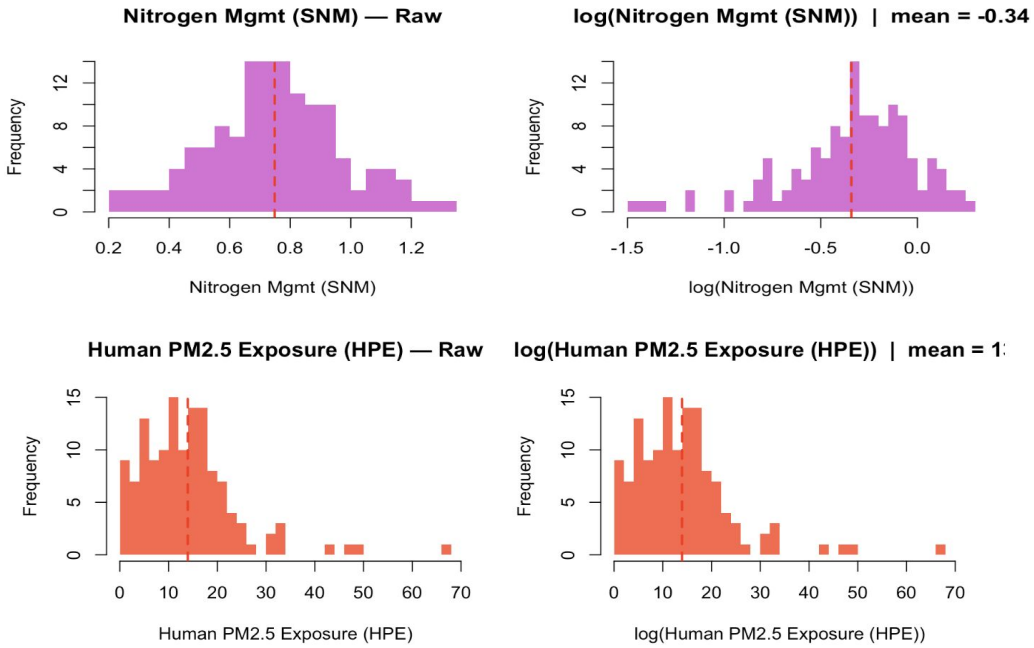
* Log transformations are base e

Model Preparation - Choosing raw vs log distributions



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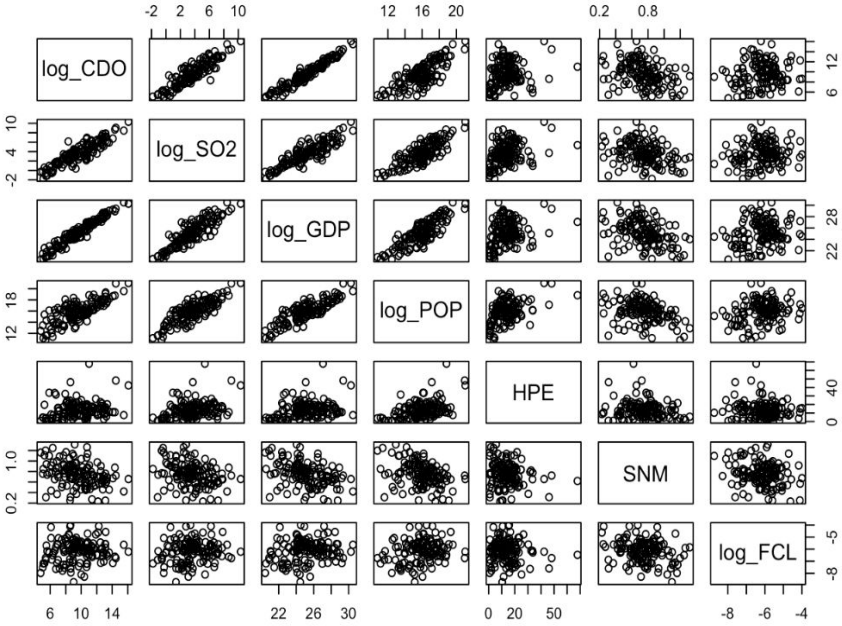
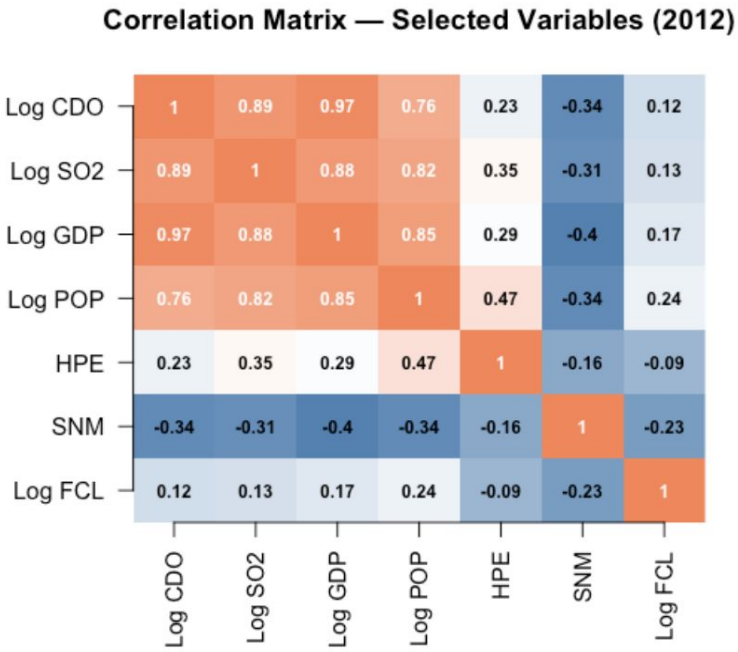
Model Preparation - Correlation Matrix





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Final Variable Selection

Category	Variable	Description	Transformations done	Decision on whether to use in final modeling
Industrial activity	CDO	CO2 emissions	Log	Yes, strongest indicator
~~Industrial activity~~	~~SO2~~	~~SO2 emissions~~	~~Log~~	Highly correlated with CO2 emissions, did not include
Economic development	GDP	Gross domestic product	Log	Yes, potential development indicator
Economic development	HPE	Anthropogenic PM2.5 exposure	Raw	Yes, pollution indicator
Economic development	SNM	Sustainable Nitrogen Management Index	Raw	Yes
Other	POP	Population	Log	Yes
Ecological damage	FCL	Forest cover loss	Log	Yes, key indicator variable
~~Ecological damage~~	~~FSS~~	~~Fish stock status~~		
~~Ecological damage~~	~~PFL~~	~~Primary forest loss~~		

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Our model interpretation approach

- Descriptive focus: interpret relationships, not prove causation
  - Focus on interpreting magnitude and sign of coefficient estimates
- Summary statistics evaluated:
  - R2
  - P-values (keeping in mind multiple comparisons)
- Generalizability matters heavily - does the model hold from 2012-2021?

Model 1

Model 1 was the simplest credible model which we defined to be

log(FCL) ~ log(CDO) + log(POP)

We believed CDO could generalize the best as a single standing variable and would set up a strong foundation for adding more later

```
Call:
lm(formula = log_FCL ~ log_CDO + log_POP, data = df_train)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.48295 -0.53589  0.04993  0.54353  2.04335 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -8.20533    0.69036 -11.886  < 2e-16 ***
log_CDO     -0.05379    0.04932  -1.091  0.27747    
log_POP      0.16101    0.06138   2.623  0.00975 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8825 on 130 degrees of freedom
Multiple R-squared:  0.06401,	Adjusted R-squared:  0.04961 
F-statistic: 4.445 on 2 and 130 DF,  p-value: 0.01357
```

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Model 2a

Model 2a is defined as

log(FCL) ~ log(CDO) + log(GDP) + log(POP)

Added GDP as a representation of economic development

GDP is highly correlated with CDO and therefore does not contribute that much unique signal from the data

```
Call:
lm(formula = log_FCL ~ log_CDO + log_GDP + log_POP, data = df_train)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.4646 -0.5562  0.0730  0.5661  2.0673 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -11.67695    2.71350  -4.303  3.3e-05 ***
log_CDO      -0.23348    0.14449  -1.616    0.109    
log_GDP       0.25189    0.19045   1.323    0.188    
log_POP       0.08908    0.08187   1.088    0.279    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8799 on 129 degrees of freedom
Multiple R-squared:  0.07654,	Adjusted R-squared:  0.05506 
F-statistic: 3.564 on 3 and 129 DF,  p-value: 0.0161
```

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Model 2b

Model 2b is defined as

log(FCL) ~ log(CDO) + log(POP) + HPE

HPE (air quality) is less correlated with emissions and may be a more orthogonal indicator of economic development

Note the improved R2 (0.10) vs. model 2 (0.05)

```
Call:
lm(formula = log_FCL ~ log_CDO + log_POP + HPE, data = df_train)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.52417 -0.52429  0.02659  0.48173  1.99944 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -9.046151   0.721783 -12.533  < 2e-16 ***
log_CDO     -0.088059   0.049039  -1.796 0.074882 .  
log_POP      0.257809   0.067179   3.838 0.000194 ***
HPE         -0.026505   0.008566  -3.094 0.002422 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8547 on 129 degrees of freedom
Multiple R-squared:  0.1287,	Adjusted R-squared:  0.1084 
F-statistic:  6.35 on 3 and 129 DF,  p-value: 0.0004755
```

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Model 2c

Model 2c is defined as

log(FCL) ~ log(CDO) + log(POP) + HPE + SNM

```
Call:
lm(formula = log_FCL ~ log_CDO + log_POP + HPE + SNM, data = df_train)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.27142 -0.49728  0.03221  0.50754  1.91733 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -8.050695   0.834754  -9.644  < 2e-16 ***
log_CDO     -0.102539   0.048686  -2.106 0.037147 *  
log_POP      0.241844   0.066494   3.637 0.000398 ***
HPE         -0.027108   0.008436  -3.213 0.001660 ** 
SNM         -0.793153   0.349230  -2.271 0.024808 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8413 on 128 degrees of freedom
Multiple R-squared:  0.1624, Adjusted R-squared:  0.1362 
F-statistic: 6.205 on 4 and 128 DF,  p-value: 0.000135
```

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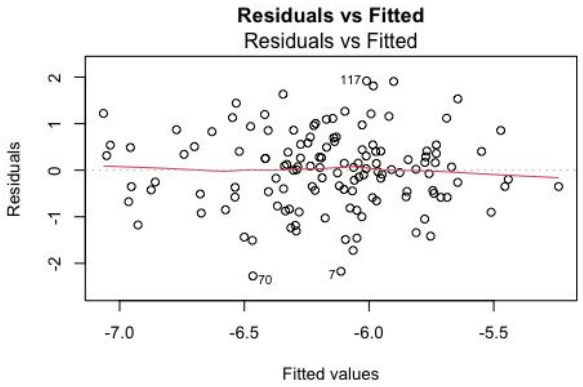
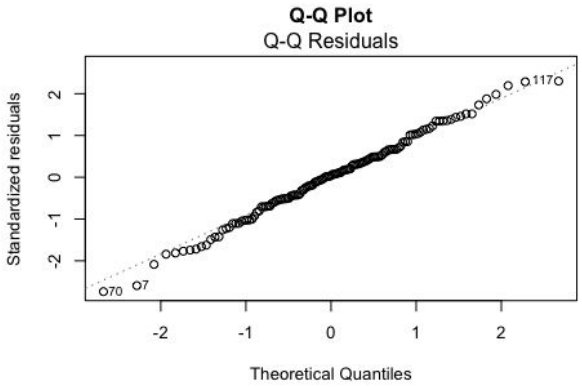
Model Interpretation - Model 2c

- Residuals appear randomly scattered with no strong pattern
- Q-Q plot is close to linear, indicating normally distributed residuals
- Low VIF values indicate no serious multicollinearity
- Non-significant Breusch-Pagan test ($p \approx 0.47$) suggests no significant heteroskedasticity






```
=== VIF (Model 2c) ===
log_CDO log_POP     HPE     SNM
  2.578   3.105   1.348   1.150

=== Breusch-Pagan Test (Model 2c) ===

        studentized Breusch-Pagan test

data:  m2c
BP = 3.52, df = 4, p-value = 0.4748
```

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Testing model 2c on another year (2021)

Model 2c (Training set, 2012)

```
Call:
lm(formula = log_FCL ~ log_CDO + log_POP + HPE + SNM, data = df_train)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.27142 -0.49728  0.03221  0.50754  1.91733 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -8.050695   0.834754  -9.644  < 2e-16 ***
log_CDO     -0.102539   0.048686  -2.106 0.037147 *  
log_POP      0.241844   0.066494   3.637 0.000398 ***
HPE         -0.027108   0.008436  -3.213 0.001660 ** 
SNM         -0.793153   0.349230  -2.271 0.024808 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8413 on 128 degrees of freedom
Multiple R-squared:  0.1624,	Adjusted R-squared:  0.1362 
F-statistic: 6.205 on 4 and 128 DF,  p-value: 0.000135
```

Model 2c (Test set, 2021)

```
Call:
lm(formula = log_FCL ~ log_CDO + log_POP + HPE + SNM, data = df_test)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.1133 -0.5154  0.0367  0.7132  3.3031 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -8.89453    1.07645  -8.263 1.61e-13 ***
log_CDO     -0.22636    0.07169  -3.158  0.00199 ** 
log_POP      0.38822    0.08974   4.326 3.04e-05 ***
HPE         -0.03073    0.01101  -2.792  0.00605 ** 
SNM         -0.79735    0.43477  -1.834  0.06900 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.088 on 127 degrees of freedom
Multiple R-squared:  0.1595,	Adjusted R-squared:  0.133 
F-statistic: 6.024 on 4 and 127 DF,  p-value: 0.00018
```

Summary statistics for train/test data:

```
RMSE (train, 2012): 0.8253
RMSE (test,  2021): 1.1078
R2   (train, 2012): 0.1624
R2   (test,  2021): 0.0942
```

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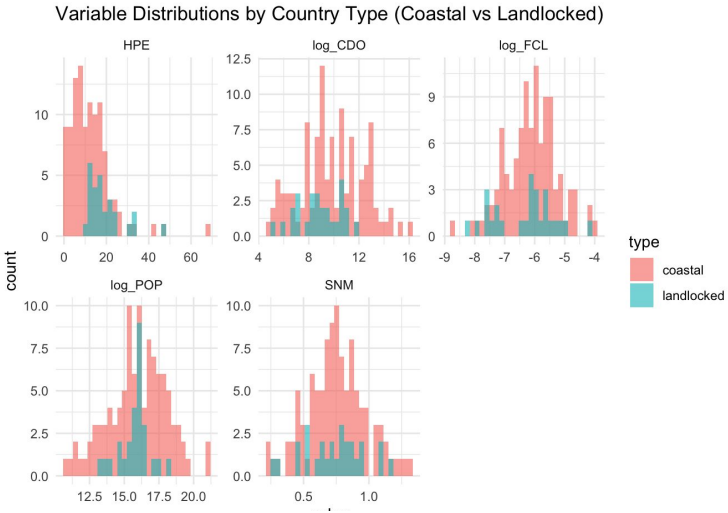
Back Up slides below

Other attempts at modeling

Attempted stratification by geographic location (coastal vs. landlocked)

log(FCL) ~ log(CDO) + log(POP) + HPE + SNM + type

Minimal improvement to the best-performing model from including this term




```
Call:
lm(formula = log_FCL ~ log_CDO + log_POP + HPE + SNM + type,
    data = df_train)

Residuals:
     Min       1Q   Median       3Q      Max
-2.28806 -0.51759  0.02895  0.53095  1.90182

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    -7.972550   0.845620  -9.428 2.54e-16 ***
log_CDO        -0.107967   0.049536  -2.180 0.031135 *
log_POP         0.241204   0.066657   3.619 0.000426 ***
HPE            -0.025423   0.008859  -2.870 0.004812 **
SNM            -0.811618   0.351238  -2.311 0.022460 *
typelandlocked -0.125219   0.196346  -0.638 0.524787
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8432 on 127 degrees of freedom
Multiple R-squared:  0.1651,    Adjusted R-squared:  0.1322
F-statistic: 5.023 on 5 and 127 DF,  p-value: 0.0003117
```

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